

Automatically Structured Interface Fields in TMI-Physics 2.2

From Quantum Possibility Distributions to Scalable Interface
Manifolds

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2026

Abstract

This paper formulates the physical layer of the Theory of Manifold Interfaces at the level of TMI-Physics 2.2. The minimal scalar interface field χ_I , introduced in earlier physical modules, is preserved as a low-dimensional amplitude projection of a fuller automatically structured interface field Ξ_I^{auto} . The central chain is: geometry defines admissible transitions; quantum theory distributes possible transitions; the primary interface provides the ontological ground of distinction; and an automatically structured interface field organizes distributed possibilities into stable structures. The paper distinguishes internal theorems of the model from physical hypotheses and experimental consequences.

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1 Purpose and Status

This document should be read as a physical module depending on TMI-Core 0.2, where automaticity is introduced as a definitional property of valid interfaces. The goal is not to prove that Ξ_I^{auto} already exists in nature. The goal is to formulate the internal physical architecture that follows if a physical interface field is admitted inside TMI-Physics.

2 Minimal Physical Chain

The minimal earlier physical chain was

$$\boxed{g \Rightarrow \text{Adm}_c(g) \Rightarrow \mathcal{H}_I(g) \Rightarrow \Psi_I \Rightarrow \text{Dist}_{\text{poss}}(\Psi_I; g) \Rightarrow \chi_I \Rightarrow \text{Struct}_I.} \quad (1)$$

In TMI-Physics 2.2 this is not discarded. It is reinterpreted as a projection of the fuller chain

$$\boxed{g \Rightarrow \text{Adm}_c(g) \Rightarrow \mathcal{H}_I(g) \Rightarrow \Psi_I \Rightarrow \text{Dist}_{\text{poss}}(\Psi_I; g) \Rightarrow \Xi_I^{\text{auto}} \Rightarrow \text{Struct}_I.} \quad (2)$$

The scalar field χ_I remains the minimal observable amplitude projection of the full field.

3 Interpretive Summary

In ordinary language, the physical layer can be stated as follows:

Geometry determines which transitions are admissible. Quantum theory distributes the possible transitions. The primary interface creates the very possibility of distinction. Therefore an automatically structured interface field can self-organize these possibilities. Coherent fields form a manifold. The manifold stabilizes as a physical structure.

4 Geometric Admissibility

Let $(\mathcal{M}, g)$ be spacetime with metric g . The domain of causally admissible transitions is

$$\text{Adm}_c(g) := \{(x, y) \in \mathcal{M} \times \mathcal{M} \mid y \in J_g^+(x)\}. \quad (3)$$

The first layer is therefore

$$\boxed{g \Rightarrow \text{Adm}_c(g).} \quad (4)$$

5 Quantum Distribution of Possibilities

Over the domain of admissible transitions we construct the Hilbert space

$$\mathcal{H}_I(g) := L^2(\text{Adm}_c(g), \mathcal{B}_g, \mu_g). \quad (5)$$

If $\Psi_I \in \mathcal{H}_I(g)$, then the state defines a distribution of possibilities:

$$\boxed{\Psi_I \Rightarrow \text{Dist}_{\text{poss}}(\Psi_I; g)}. \quad (6)$$

6 Axiomatic Source of Automaticity

From TMI-Core 0.2 we have

$$\boxed{N_0 \xrightarrow{I_0} \Delta_0 \rightarrow (A, \neg A)}. \quad (7)$$

$$\boxed{\text{Valid}(I) \Rightarrow \text{Auto}(I)}. \quad (8)$$

$$\boxed{\text{Auto}(I) \equiv (\text{Input}(I) \wedge \text{Adm}(I) \wedge \text{Dir}(I) \Rightarrow \text{AutoAct}(I))}. \quad (9)$$

If a physical field is an interface field, then

$$\boxed{\Xi_I \in \mathfrak{I}_{\text{phys}} \Rightarrow \text{Auto}(\Xi_I) \Rightarrow \Xi_I^{\text{auto}}}. \quad (10)$$

7 Automatically Structured Interface Field

Definition 1. An automatically structured interface field is an operator field

$$\Xi_I^{\text{auto}} \in \Gamma(\text{End}(\mathcal{H}_I)), \quad (11)$$

with an internal law of self-updating:

$$\partial_\tau \Xi_I^{\text{auto}} = \mathcal{A}_I(\Xi_I^{\text{auto}}, g, \Psi_I, \text{Adm}_c(g), \text{Dist}_{\text{poss}}(\Psi_I; g)). \quad (12)$$

The operator $\mathcal{A}_I$ is the internal auto-structuring rule. It does not represent an external controller, but a law internal to the interface-field dynamics.

8 Internal Structuring Theorem

Theorem 1. If $\Xi_I^{\text{auto}} \in \Gamma(\text{End}(\mathcal{H}_I))$, its dynamics is autonomous, closed in the admissible space, and there exists a stability functional $\mathcal{L}_I \geq 0$ such that

$$\frac{d}{d\tau} \mathcal{L}_I[\Xi_I^{\text{auto}}] \leq 0, \quad (13)$$

then

$$\boxed{\Xi_I^{\text{auto}} \Rightarrow \text{Struct}_I}. \quad (14)$$

Proof. The field Ξ_I^{auto} acts on the possibility space $\mathcal{H}_I$. Autonomy excludes an external controlling operator, closure preserves admissibility, and the non-increase of $\mathcal{L}_I$ defines motion toward a stable regime. This regime is denoted by $\text{Struct}_I := \text{Attr}(\mathcal{A}_I)$. Therefore $\Xi_I^{\text{auto}} \Rightarrow \text{Struct}_I$. $\square$

9 Projections

The full field admits the projection chain

$$\boxed{\Xi_I^{\text{auto}} \rightarrow \Xi_I \rightarrow \chi_I.} \quad (15)$$

The scalar projection can be written as

$$\boxed{\chi_I^2(x) = \frac{1}{N_I} \text{Tr}_I [(\Xi_I^{\text{auto}}(x))^\dagger \Xi_I^{\text{auto}}(x)] .} \quad (16)$$

The stable interface structure is represented by

$$\boxed{\text{Struct}_I = \mathcal{S}_I(g, \Psi_I, \Xi_I^{\text{auto}}, \text{Adm}_c(g), \text{Dist}_{\text{poss}}(\Psi_I; g)).} \quad (17)$$

10 Scaling

A local automatically structured field scales into a network and then into a manifold:

$$\boxed{\Xi_I^{\text{auto}} \rightarrow \{\Xi_{I,k}^{\text{auto}}\}_{k \in K} \rightarrow \mathcal{N}_\Xi \rightarrow \mathcal{M}_\Xi^{\text{auto}} \rightarrow \text{Struct}_I(\mathcal{M}).} \quad (18)$$

The manifold of automatically structured interface fields is

$$\boxed{\mathcal{M}_\Xi^{\text{auto}} = (V_\Xi, E_\Xi, O_\Xi, \mathcal{A}_\Xi, \Pi_{\text{str}}, \Pi_{\text{amp}}, \Pi_{\text{struct}}).} \quad (19)$$

Here V_Ξ is the set of local automatically structured fields, E_Ξ is the set of relations between them, O_Ξ is the order of coherence, $\mathcal{A}_\Xi$ is the global law of auto-structuring, and $\Pi_{\text{str}}, \Pi_{\text{amp}}, \Pi_{\text{struct}}$ are structural, amplitude, and stability projections.

11 Coherence and Synchronization

For linked fields, define

$$\text{Coh}_\Xi(k, l) = 1 - \text{Loss}_\Xi(\Xi_{I,k}^{\text{auto}}, \Xi_{I,l}^{\text{auto}}). \quad (20)$$

The manifold condition is

$$\text{Coh}_\Xi(\mathcal{M}_\Xi^{\text{auto}}) \rightarrow 1, \quad \text{Sync}_\Xi(\mathcal{M}_\Xi^{\text{auto}}) \rightarrow 1. \quad (21)$$

This condition expresses that local fields should not merely coexist; they should become mutually coherent and synchronized enough to form a stable manifold.

12 Computational Support and Engineering Projection

The computational validation module tests an engineering projection of this architecture in quantum-control models:

$$I_Q^{coh-auto} \Rightarrow T_2^{eff} \uparrow \Rightarrow P_{success} \uparrow. \quad (22)$$

The extended test line also introduces a safety boundary:

$$\text{Auto}(I) \not\Rightarrow \text{Safe}(I), \quad (23)$$

with the engineering recovery principle

$$\text{Auto}(I) + \text{SafetyLayer} \Rightarrow \text{SaferAdaptiveControl}. \quad (24)$$

This computational support does not prove the existence of Ξ_I^{auto} in nature. It supports the internal engineering idea that an automatic interface can improve observable quantum-control metrics under appropriate feedback and safety conditions.

13 Status of Claims

- Definitions: $\text{Valid}(I) \Rightarrow \text{Auto}(I)$.
- Internal consequences: $\Xi_I^{\text{auto}} \wedge \text{Stab}(\Xi_I^{\text{auto}}) \Rightarrow \text{Struct}_I$.
- Physical hypotheses: Ξ_I^{auto} exists as a physical mechanism.
- Experiments: $T_2^{TMI} > T_2^{\text{base}}$ in a quantum implementation.

14 Conclusion

TMI-Physics 2.2 formulates the physical layer as a projection of the updated core of automatic interfaces. The scalar field χ_I is preserved as the minimal observable projection, while the full form Ξ_I^{auto} describes automatic structuring of distributed possibilities and scaling toward an interface manifold.

References

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